

Arithmetic Operations and Scientific Calculations

1 The calculator Package

1.1 Basic Arithmetic

Storing Numbers We copy 0.5 to `\half` So, 0.5
New half is 0.5

Addition Add (8, 7) to get 15

Subtraction Sub (8 from 10) to get 2

Multiplication Multiply (7.5, 6.3) to get 47.25002

Division Divide (80, by 7) to get 11.42857

Maximum The Max of (7, 9) is . So, 9

Minimum The Min of (5, 3) is . So, 3

Simplifying Fractions The numerator and denominator are (5, 8)

Result as a New Macro My new Macro now holds 16

1.2 Using in New Commands

You still in bound, $1 < 50$

You're Free, $27 > 25$

1.3 The Scope of Result Variables

`\eight` is 8

At First `\eight` is 8

After adding one: `\eight` becomes 9!!

`\eight` is 8

1.4 Working with Lengths

Length Addition The length `height` is 122.26999pt

Length Subtraction Now, `height` is 36.9055pt

Length Division One Inch is equal to 2.54

1.5 Operations on Vectors

Assigning 2-D Vectors My new vector is (3, 4)

Assigning 3-D Vectors Making it 3-D (3, 4, 2)

Vector Addition Adding the new vector to itself. So, (6, 8)

Scalar-Vector Product So, it becomes (6, 8)

1.5.1 More than Three Elements

Adding (1, 2, 3, 4) to itself gives (2, 4, 6, 7.99999)

While the result must be: (2, 4, 6, 8)

1.6 Operations on Matrices

Assigning 2 x 2 Matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Assigning 3 x 3 Matrix $\begin{bmatrix} 2 & 1 & 4 \\ 3 & -5 & 5 \\ 2 & -7 & 4 \end{bmatrix}$

Transpose a Matrix $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

Matrix Addition $\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$

Multiply by Scalar $\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$

2 The calculus Package

2.1 Built-in Functions

Square Function If we have an input $t = 4$ and $f(t) = t^2$, and it's derivative is $f'(t) = 2t$, then: $f(4) = 16$, and $f'(4) = 8$.

Cube Function If we have an input $t = 2$ and $f(t) = t^3$, and it's derivative is $f'(t) = 3t^2$, then: $f(2) = 8$, and $f'(2) = 12$.

Square Root If we have an input $t = 16$ and $f(t) = \sqrt{t}$, and it's derivative is $f'(t) = \frac{1}{2\sqrt{t}}$, then: $f(16) = 4$, and $f'(16) = 0.125$

2.2 Defining New Functions via Operations

Summing Two Functions We can define $F(t) = \sqrt{t} + \sin t$ using the `\SUMfunction`. So that we have:

- $F(4) = \sqrt{4} + \sin 4$ equal to 1.2432
- $F'(4) = \frac{1}{2\sqrt{4}} + \cos 4$ equal to -0.40364

Some Other Operations that work the same way:

- `\SUBTRACTfunction{\f1}{\f2}{\newSubFun}` $F(t) = f1 - f2$
- `\PRODUCTfunction{\f1}{\f2}{\newProdFun}` $F(t) = f1 * f2$
- `\SCALEfunction{num}{\func}{\ScFunc}` $F(t) = num * func$

2.3 Example: Linear Combination

The linear combination of two functions $f(x)$ and $g(x)$ is given by:

$$h(x) = a \cdot f(x) + b \cdot g(x)$$

Compose and Evaluate: $F(t) = 2e^{\cos t} - e^{-t} \sin t$

- $F(4) = 2e^{\cos 4} - e^{-4} \sin 4$, which is 1.05421
- $F'(t) = -2e^{\cos t} \sin t - e^{-t} \cos t + e^{-t} \sin t$, for $t = 4$ we have 0.78543

2.4 Defining Polynomial Functions

Linear Polynomial $p(t) = a + b t$

at $t = 2$, it evaluates to $p(2) = 10$, $p'(2) = 3$

Quadratic Polynomial $p(t) = a + b t + c t^2$

at $t = 2$, it evaluates to $p(2) = 14$, $p'(2) = 7$

Cubic Polynomial $p(t) = a + b t + c t^2 + d t^3$

at $t = 2$, it evaluates to $p(2) = 26$, $p'(2) = 23$

Of course this whole setup can be abstracted inside a new control sequence.

Evaluating $p(t) = a + b t + c t^2$ at $t = 4$ gives:- $p(4) = 57$, and $p'(4) = 26$

2.5 Vector Valued Functions

$$\mathbf{F}(t) = \begin{bmatrix} t^2 \\ t^3 \end{bmatrix}$$

If parameter/variable $t = 2$:

$$F(2) = \begin{bmatrix} 4 \rightarrow 2^2 \\ 8 \rightarrow 2^3 \end{bmatrix}$$

and

$$F'(2) = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$$

2.6 Combining Two Polynomials

$$\mathbf{P}(t) = \begin{pmatrix} a + b t \\ a + b t + c t^2 \end{pmatrix}$$

Evaluating at $a = 1, b = 2, c = 3$, and $t = 4$, gives:

$$P(4) = \begin{pmatrix} 178 \\ 7038 \end{pmatrix}$$

and

$$P'(4) = \begin{pmatrix} 12 \\ 992 \end{pmatrix}$$

2.6.1 Defining Custom Vector Functions

$$\mathbf{V}(t) = \begin{bmatrix} X(t) = t^2 \\ Y(t) = t^3 \end{bmatrix}$$

$$V(2) = \begin{bmatrix} 4 \\ 8 \end{bmatrix}, V'(2) = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$$

2.7 Defining Low Level New Functions

Defining $F(t) = t^2 + t^3$ which has a derivative of $F'(t) = 2t + 3t^2$

$$F(t) = 12, F'(t) = 16$$